

Kun-Yu Lee

Evanston, IL · kun-yulee2027@u.northwestern.edu · quentinkunyu.com · GitHub · LinkedIn

Research Interests

Evaluating how LLM-based recommenders decide under uncertainty: when a missing preference is worth a clarifying question, which options they retrieve and rank, and whether this behavior can be reproducibly audited.

Education

Northwestern University

Sep 2025 – Expected Dec 2026

M.S. in Machine Learning and Data Science

Evanston, IL

- GPA: 3.94/4.00. Coursework: Machine Learning I–II, Deep Learning, Natural Language Processing, Generative AI, Databases and Information Retrieval, Interpretable Machine Learning

University of Nebraska–Lincoln

Aug 2021 – Dec 2024

B.S. in Computer Science and B.A. in Data Science; Minor in Mathematics

Lincoln, NE

- GPA: 3.91/4.00; B.S. with High Distinction, B.A. with Distinction
- Coursework: Linear Algebra, Numerical Analysis, Statistics and Applications (probability and inference), Automata, Computation, and Formal Languages

Publications and Manuscripts

- [1] **Kun-Yu Lee** and Edward C. Malthouse. “Evaluating Whether LLM Recommendations Respond to the Decision Value of Missing Preferences.” *Under review at the European Conference on Information Retrieval (ECIR 2027), short paper track, October 2026.*
- [2] Edward C. Malthouse, **Kun-Yu Lee**, Jing Yang, Sanchary Pal, and Xueyan Feng. “The Silent Gatekeeper: A Framework for Measuring, Explaining, and Diagnosing Brand Visibility in LLM Recommendations.” *Under review at the Journal of Advertising Research (JAR), 2026.*
- [3] Edward C. Malthouse, **Kun-Yu Lee**, Jing Yang, Sanchary Pal, and Xueyan Feng. “Evaluating Brand Retrieval and Ranking in Open-Ended LLM Recommendations.” *Under review at the European Conference on Information Retrieval (ECIR 2027).* Extended version: “Evaluating Brand Retrieval and Ranking in Large Language Model Recommendations,” *arXiv preprint arXiv:2609.16304*, 2026.
- [4] Jing Yang, **Kun-Yu Lee**, Sanchary Pal, Xueyan Feng, and Edward C. Malthouse. “The Machine Brand Image: Evaluating LLM Recommendations Across Prompts and Models.” *Under review at the American Academy of Advertising (AAA) 2027 Conference.*

Research Experience

Graduate Research Assistant, Northwestern University

Jan 2026 – Present

Advisors: Edward C. Malthouse and Jing Yang (Boston University)

Evanston, IL

LLM Recommendations and the Decision Value of Missing Preferences

First author; Paper [1]

- *Finding:* LLMs raised missing preferences mainly when consumers mentioned them, not when the recommendation depended on them: 70–93% of responses addressed open preferences stated as undecided, but only 10–27% did when the same preferences went unmentioned, despite 97–100% accuracy with complete preferences.
- *Design:* Built a benchmark of 72 hotel-choice scenarios (61 cities) from HotelRec ratings with an exact oracle for whether a consumer’s undecided preferences can change the recommendation; paired requests (“book one” vs. “the best three”) make the same missing preference decision-relevant in one condition but not the other, so clarifying-question behavior is judged by its decision value.
- *Evaluation:* 2,160 queries to GPT-5.6 Luna, Gemini 3.6 Flash, and Claude Sonnet 5; response types coded by three LLM annotators from providers outside the study and validated against blind human coding ($\kappa = 0.98$, $n = 100$); 95% scenario-cluster bootstrap intervals.

Auditing Brand Recommendations in LLMs

Second author; Papers [2]–[4]

- *Framework*: Audits open-ended LLM brand recommendations as a stochastic retrieval-and-ranking process: six commercial LLMs, five categories, repeated stateless sampling, and BRP@ k / MRR@ k for recommendation prevalence and prominence.
- *Role*: Designed and ran the evaluation experiments and analyzed the results; contributed to the research framing and co-wrote the manuscripts.
- *Findings*: Category-only queries omit established brands (e.g., Craftsman, BRP@5 = 0%), and search interest is the most consistent predictor of recommendation prominence; needs-based NDCG varies widely across brand-positioning dimensions.
- *Machine brand image*: From 17,821 LLM justifications coded into eight association types, price tier is the strongest marketplace correlate, while advertising spending shows no association.

Earlier Research, University of Nebraska–Lincoln

Nov 2023 – May 2025

Undergraduate Research Assistant

Lincoln, NE

- *Viral genomics* (Advisor: Qiuming Yao): built an interactive tool to study how mutation patterns vary across virus lineages, regions, and time.
- *EV adoption* (Advisor: Jason Fraser Hawkins): modeled how infrastructure, geography, and time relate to consumer EV adoption using interpretable regression and tree-based models.

Selected Projects

Interpretable LLM Agent for Structured Scouting and Ranking

Hackathon Third Place, Nov 2025

- Built an agentic workflow that applies explicit normalization, constraint, and conflict-resolution rules before ranking candidates; generated rationales for human review and tested ranking stability.

Honors and Awards

Third Place, Northwestern MLDS AI Hackathon

Nov 2025

Dean’s List, University of Nebraska–Lincoln

2021–2024

Teaching

Learning Assistant, Department of Mathematics, University of Nebraska–Lincoln

Aug 2023 – Dec 2023

Undergraduate mathematics

Lincoln, NE

- Led weekly recitations and office hours and helped students build mathematical reasoning for modeling and data analysis.

Industry Experience

Financial Trading Assistant, CTBC Financial Holding

Jun 2023 – Aug 2023

Market-data feature engineering and visualization

Taipei, Taiwan

- Engineered real-time quantitative features (price, volatility) from high-frequency Bloomberg data and built dashboards used by traders and analysts.